# Linear Algebra: linear functionals

Chong Jongwon

June 30, 2026

## Contents

# 1 Dual Space

Let $V$ be a vector space over a field $F$.

**Definition 1.1.** A linear map $f : V \to F$ is called a *linear functional* on $V$.

**Example.** Let $F[x]$ be the set of polynomials over $F$. For each $\alpha \in F$, define

$$L_\alpha : F[x] \to F : p(x) \mapsto p(\alpha)$$

This is called the 'evaluation' at $\alpha$. Clearly, it is a linear functional since

$$L_\alpha(c \cdot p(x) + q(x)) = c \cdot p(\alpha) + q(\alpha) = c \cdot L_\alpha(p(x)) + L_\alpha(q(x))$$

**Example.** Let $C([a, b])$ be the space of continuous real-valued functions on $[a, b]$. Define

$$L : C([a, b]) \to \mathbb{R} : f \mapsto \int_a^b f(t)\, dt$$

$L$ is clearly a linear functional by the linearity of the integral.

**Definition 1.2.** The collection of all linear functionals on $V$ is called the *dual space* of $V$ and is denoted as $V^*$.

**Theorem 1.3.** Let $V$ be a finite-dimensional vector space over $F$, and let $\beta = \{\alpha_1, \dots, \alpha_n\}$ be a basis for $V$.
Then, there exists a unique *dual basis* $\beta^* = \{f_1, \dots, f_n\}$ for $V^*$ such that $f_i(\alpha_j) = \delta_{ij}$.
Furthermore, for each linear functional $f \in V^*$,

$$f = \sum_{i=1}^{n} f(\alpha_i) f_i$$

and for each vector $\alpha \in V$,

$$\alpha = \sum_{i=1}^{n} f_i(\alpha)\alpha_i$$

*Proof.* By the property of basis, for any $\alpha \in V$, there exist unique scalars $a_1, \dots, a_n \in F$ such that

$$\alpha = a_1\alpha_1 + \dots + a_n\alpha_n$$

For each $1 \le i \le n$, define $f_i : V \to F$ by $f_i(\alpha) = a_i$ (or $f_i(\alpha) = ([\alpha]_\beta)_i$).
Then $f_i(\alpha_j) = \delta_{ij}$. Since $\dim V = \dim V^*$, it is enough to show that the set $\beta^* = \{f_1, \dots, f_n\}$ is linearly independent.
Suppose that for some $c_i \in F$, $\sum_{i=1}^n c_i f_i = 0$. Then for each $1 \le j \le n$,

$$\left(\sum_{i=1}^{n} c_i f_i\right)(\alpha_j) = \sum_{i=1}^{n} c_i f_i(\alpha_j) = c_j = 0$$

Thus, $\beta^*$ is linearly independent and forms a basis for $V^*$.
The second assertion can be proved as follows: if $\alpha \in V$ is represented as $\alpha = \sum_{i=1}^n x_i\alpha_i$ where $x_i \in F$, then for each $1 \le j \le n$:

$$f_j(\alpha) = f_j\left(\sum_{i=1}^{n} x_i\alpha_i\right) = \sum_{i=1}^{n} x_i f_j(\alpha_i) = x_j$$

Substituting $x_j$ back into the representation of $\alpha$, we obtain $\alpha = \sum_{i=1}^n f_i(\alpha)\alpha_i$. □

## Application: Lagrange Interpolation

Let $P_n(F) = \{p \in F[x] : \deg p \leq n\}$, and let $t_0, t_1, \ldots, t_n \in F$ be distinct elements.
For each $0 \leq i \leq n$, define the linear functional $L_i : P_n(F) \to F$ by $L_i(p) = p(t_i)$.
These $\{L_0, L_1, \ldots, L_n\}$ are linearly independent. Suppose that

$$c_0 L_0 + c_1 L_1 + \cdots + c_n L_n = 0 \quad \cdots (*)$$

where $c_i \in F$. By substituting the standard basis $\{1, x, x^2, \ldots, x^n\}$ into $(*)$, we obtain a system of equations:

$$\begin{cases} c_0 + c_1 + \cdots + c_n = 0 \\ c_0 t_0 + c_1 t_1 + \cdots + c_n t_n = 0 \\ \qquad\vdots \\ c_0 t_0^n + c_1 t_1^n + \cdots + c_n t_n^n = 0 \end{cases} \implies \begin{bmatrix} 1 & 1 & \cdots & 1 \\ t_0 & t_1 & \cdots & t_n \\ \vdots & \vdots & \ddots & \vdots \\ t_0^n & t_1^n & \cdots & t_n^n \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The matrix above is the **Vandermonde Matrix**, which is invertible since $t_i$ are distinct. Thus $c_0 = c_1 = \cdots = c_n = 0$.
More precisely, define a linear map

$$T : P_n(F) \to F^{n+1} : p(x) \mapsto (p(t_0), \ldots, p(t_n))$$

For $\beta = \{1, x, \ldots, x^n\}$, $[T]_\beta$ is the Vandermonde Matrix.
If we choose a different basis $\beta' = \{1, (x - t_0), (x - t_0)(x - t_1), \ldots, \prod_{j=0}^{n-1}(x - t_j)\}$, then $[T]_{\beta'}$ becomes:

$$\det[T]_{\beta'} = \det \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & t_1 - t_0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & t_n - t_0 & \cdots & \prod_{j=0}^{n-1}(t_n - t_j) \end{bmatrix} = \prod_{1 \leq i < j \leq n} (t_j - t_i) \neq 0$$

Since the determinant is invariant with respects to the similar matrix, it is proved.

> **Theorem 1.4.** $\{L_0, \ldots, L_n\}$ is a dual basis for $P_n(F)^*$ with respect to the basis $\{P_0(x), \ldots, P_n(x)\}$ for $P_n(F)$, where
>
> $$P_i(x) = \prod_{j \neq i} \frac{x - t_j}{t_i - t_j}$$
>
> Since $L_j(P_i) = P_i(t_j) = \delta_{ij}$, for any $p(x) \in P_n(F)$, we have:
>
> $$p(x) = \sum_{i=0}^{n} L_i(p) P_i(x) = \sum_{i=0}^{n} p(t_i) P_i(x)$$
>
> Furthermore, by the linearity of the integral, $\displaystyle\int_a^b p(x)dx = \sum_{i=0}^{n} p(t_i) \int_a^b P_i(x)dx$.

## 2 Double Dual

*Question: Is every basis for $V^*$ a dual of some basis for $V$?*

**Lemma 2.1.** Let $V$ be a finite dimensional vector space over $F$. For each $\alpha \in V$, define

$$\widehat{\alpha} : V^* \to F : f \mapsto f(\alpha)$$

Then, it is linear functional on $V^*$. Moreover, if $\widehat{\alpha}(f) = 0$ for any $f \in V^*$, then $\alpha = 0$.

Note: the last assertion can be said: $(V^*)^0 = \{0\}$.

*Proof.* Let $f, g \in V^*$ and $c \in F$ be given. Then,

$$\widehat{\alpha}(cf + g) = (cf + g)(\alpha) = cf(\alpha) + g(\alpha) = c\widehat{\alpha}(f) + \widehat{\alpha}(g)$$

Thus proved. To show the second assertion, let $\alpha \neq 0$. Then, there exists a basis $\beta$ for $V$ containing $\alpha$. Then, there is a dual basis $\beta^*$ for $V^*$. Take $f_\alpha \in \beta^*$ corresponding to $\alpha$. Then, $f_\alpha(\alpha) = 1 \neq 0$. Thus proved. □

**Theorem 2.2.** Let $V$ be a finite dimensional vector space over $F$. Then, the map

$$V \to V^{**} : \alpha \mapsto \widehat{\alpha}$$

is an Isomorphism.

*Proof.* Let $\alpha, \beta \in V$ and $c \in F$, and for any $f \in V^*$,

$$(\widehat{c\alpha + \beta})(f) = f(c\alpha + \beta) = cf(\alpha) + f(\beta) = c\widehat{\alpha}(f) + \widehat{\beta}(f) = (c\widehat{\alpha} + \widehat{\beta})(f) = c\widehat{\alpha}(f) + \widehat{\beta}(f)$$

Moreover, by the Lemma, the kernel of the map is trivial. Thus injective, and surjective because:

$$\dim V^{**} = \dim V^* = \dim V$$

□

From the Theorem above, we obtain that:

If $\widehat{\alpha} \in V^{**}$, then there exists a unique $\alpha \in V$ such that for all $f \in V^*$, $\widehat{\alpha}(f) = f(\alpha)$.

We can restate this result as:

**Corollary 2.3.** Let $V$ be a finite dimensional vector space over $F$.
Every basis for $V^*$ is the dual of some basis for $V$.

*Proof.* Let $\{f_1, \ldots, f_n\}$ be a basis for $V^*$. Then, there is a dual basis $\{\widehat{\alpha_1}, \ldots, \widehat{\alpha_n}\}$ for $V^{**}$ such that $\widehat{\alpha}_i(f_j) = \delta_{ij}$. By the above corollary, for each $\widehat{\alpha}_i \in V^{**}$, there exists a unique $\alpha_i \in V$ such that $\widehat{\alpha}_i(f) = f(\alpha_i)$ for all $f \in V^*$. Now, $\widehat{\alpha}_i(f_j) = f_j(\alpha_i) = \delta_{ji}$, the claim is proved. □

**Theorem 2.4.** Let $V$ be a finite-dimensional vector space, and let $S \subset V$ be a subset. Then,

$$S^{00} = \widehat{\langle S \rangle}$$

where $\widehat{\langle S \rangle}$ is natural isomorphic copy of $\langle S \rangle$ in $V^{**}$.

## 3 Transpose

**Theorem 3.1.** Let $V$ and $W$ be vector spaces over $F$, and let $T: V \to W$ be a linear map. Then,

$$T^t : W^* \to V^* : g \mapsto g \circ T$$

is a linear map.
Furthermore, if $V$ and $W$ have finite dimensional with ordered basis $\beta$ and $\gamma$ for $V$ and $W$, respectively, then

$$([T]_\beta^\gamma)^t = [T^t]_{\gamma^*}^{\beta^*}$$

*Proof.* Since the composition of two linear maps is linear, the map $T^t$ is well-defined. Given $f, g \in W^*$ and $c \in F$,

$$T^t(cf + g) = (cf + g) \circ T = cf \circ T + g \circ T = cT^t(f) + T^t(g)$$

Thus, the first assertion is proved.
Now, suppose that $V$ and $W$ have finite dimensonal. Let $\beta = \{x_1, \ldots, x_n\}$ and $\gamma = \{y_1, \ldots, y_m\}$.
Then, there are dual basis for $V^*$, $W^*$

$$\beta^* = \{f_1, \ldots, f_n\}, \ \ \gamma^* = \{g_1, \ldots, g_m\}$$

respectively. Observe that: for each $1 \le j \le m$,

$$T^t(g_j) = g_j \circ T = \sum_{s=1}^{n} (g_j \circ T)(x_s) f_s$$

Meanwhile, denote $A = [T]_\beta^\gamma$. Then,

$$(g_j \circ T)(x_i) = g_j(T(x_i)) = g_j\left(\sum_{k=1}^{m} A_{ki} y_k\right) = \sum_{k=1}^{m} A_{ki} g_j(y_k) = \sum_{k=1}^{m} A_{ki} \delta_{jk} = A_{ji}$$

Therefore,

$$[T^t]_{\gamma^*}^{\beta^*} = \left[[T^t(g_1)]_{\beta^*} \quad \cdots * [T^t(g_m)]_{\beta^*}\right] = A^t = ([T]_\beta^\gamma)^t$$

$\square$

**Theorem 3.2.** Let $F$ be a field, and let $T : F^n \to F^m$ be a map. A map $T$ is linear map *if and only if*:

There exist $T_1, \ldots, T_m \in (F^n)^*$ such that for all $x \in F^n$, $T(x) = (T_1(x), \ldots, T_m(x))$

*Proof.* Let $\{g_1, \ldots, g_m\}$ be a dual basis for $F^m$ with respects to standard basis for $F^m$. For each $1 \le i \le m$, define

$$T_i = T^t(g_i)$$

Let $x \in V$ be given. Then, for each $1 \le i \le m$,

$$T_i(x) = T^t(g_i)(x) = (g_i \circ T)(x) = g_i(T(x)) = ([T(x)]_\gamma)_i$$

Thus proved. The converse is trivial. $\square$

## 4 Hyperspace

**Lemma 4.1.** Let $f, g : V \to F$ be linear functionals on $V$.

$$g = c \cdot f \text{ for some } c \in F \textit{ if and only if } \ker f \leq \ker g.$$

*Proof.* If $f = 0$, then it is trivial. Assume $f \neq 0$. Then $\ker f$ is a hyperspace of $V$.
Let $\alpha \in V$ such that $f(\alpha) \neq 0$. Put $c = g(\alpha)/f(\alpha)$.
Then, $g - cf$ is a linear functional such that $\ker f \leq \ker(g - cf)$ and $\alpha \in \ker(g - cf)$. Thus $\ker(g - cf) = V$, or $g - cf = 0$. □

**Theorem 4.2.** Let $g, f_1, \dots, f_r$ be linear functionals on $V$. Then,

$$g \text{ is a linear combination of } \{f_1, \dots, f_r\} \textit{ if and only if } \bigcap_{i=1}^{r} \ker f_i \leq \ker g.$$

*Proof.* If $g = \sum_{i=1}^{r} c_i f_i$ for some $c_i \in F$, then for any $\alpha \in \bigcap_{i=1}^{r} \ker f_i$, $g(\alpha) = \sum_{i=1}^{r} c_i f_i(\alpha) = 0$.
Conversely, we shall use the mathematical induction for $r \in \mathbb{N}$.
By the previous Lemma, if $r = 1$, it is proved. Suppose that it is held for $r = k - 1$.
Let $f_1, \dots, f_k$ be linear functionals on $V$ such that $\bigcap_{i=1}^{k} \ker f_i \leq \ker g$.
Let $g, f_1, \dots, f_{k-1}$ be restricted by $\ker f_k$. That is,

$$f_i|_{\ker f_k} : \ker f_k \to F : \alpha \mapsto f_i(\alpha) \quad (i = 0, \dots, k-1, \text{ where } f_0 = g)$$

And, let $\alpha \in \ker f_k$ such that for all $1 \leq i \leq k - 1$, $f_i(\alpha) = 0$.
That is, $\alpha \in \bigcap_{i=1}^{k} \ker f_i$ and therefore $g|_{\ker f_k}(\alpha) = 0$ by the hypothesis. By the assumption of the induction,

$$g|_{\ker f_k} = \sum_{i=1}^{k-1} c_i f_i|_{\ker f_k} \text{ for some } c_i \in F.$$

Put $h = g - \sum_{i=1}^{k-1} c_i f_i$. Now, for any $\alpha \in \ker f_k$, $h(\alpha) = 0$, or $\ker f_k \leq \ker h$.
The previous Lemma gives that $h = c_k f_k$ for some $c_k \in F$. Now, $g = \sum_{i=1}^{k} c_i f_i$. □

**Theorem 4.3.** Let $V$ be a vector space over $F$.
For any non-zero linear functional $f \in V^*$, $\ker f$ is a hyperspace of $V$.
Conversely, every hyperspace in $V$ is the kernel for some linear functional on $V$.

*Proof.* Let $f \in V^*$ be a non-zero linear functional. Since $f$ is non-zero, choose $\alpha \in V \setminus \ker f$ (or $f(\alpha) \neq 0$).
We shall show that $V = \ker f + \langle \alpha \rangle$. Let $\beta \in V$ be given. Put $c = f(\beta)/f(\alpha)$. (It can be defined because $f(\alpha) \neq 0$).
Then, $f(\beta - c\alpha) = f(\beta) - cf(\alpha) = 0$, thus $\beta - c\alpha \in \ker f$, or $\beta \in \ker f + \langle \alpha \rangle$.
Conversely, let $N \leq V$ be a hyperspace. Let $\alpha \in V \setminus N$. Since $N$ is a hyperspace, $N + \langle \alpha \rangle = V$. Now, for each $\beta \in V$,

$$\beta = \gamma + c\alpha \text{ for some } \gamma \in N, \; c \in F.$$

This $\gamma$ and $c$ are uniquely determined because $N \cap \langle \alpha \rangle = \{0\}$. Define $g : V \to F$ as $g(\beta) = c$. Then, $\ker g = N$. □

# 5 Annihilators

**Definition 5.1.** Let $V$ be a vector space over $F$, and let $S \subseteq V$ be a subset. The **annihilator** of $S$ is the set

$$S^0 \overset{\text{def}}{=} \{f \in V^* \mid f(\alpha) = 0 \text{ for all } \alpha \in S\}.$$

Note: $f \in S^0 \iff S \subseteq \ker f$.

**Theorem 5.2.** Let $V$ be a finite-dimensional vector space, and let $W \leq V$ be a subspace. Then

$$\dim W + \dim W^0 = \dim V.$$

*Proof.* Clearly, $W^0 \leq V^*$. Let $\{\alpha_1, \ldots, \alpha_k\}$ be a basis for $W$.
Choose $\alpha_{k+1}, \ldots, \alpha_n \in V$ such that $\{\alpha_1, \ldots, \alpha_n\}$ is a basis for $V$.
Let $\{f_1, \ldots, f_n\}$ be the dual basis for $V^*$ corresponding to $\{\alpha_1, \ldots, \alpha_n\}$.
For $k+1 \leq i \leq n$, we have $f_i(\alpha_j) = 0$ for all $1 \leq j \leq k$. Thus, $f_i(\alpha) = 0$ for all $\alpha \in W$, which implies $\{f_{k+1}, \ldots, f_n\} \subseteq W^0$.
Clearly, this set is linearly independent. To show that it spans $W^0$, let $f \in W^0 \leq V^*$ be given. Then

$$f = \sum_{i=1}^{n} f(\alpha_i) f_i = \sum_{i=k+1}^{n} f(\alpha_i) f_i \in \text{span}\{f_{k+1}, \ldots, f_n\}$$

since $f(\alpha_i) = 0$ for $1 \leq i \leq k$. Thus, $\dim W^0 = n - k = \dim V - \dim W$. □

**Corollary 5.3.** If $W \leq V$ with $\dim W = k < n = \dim V$, then

$$W = \ker f_{k+1} \cap \cdots \cap \ker f_n$$

where $f_i$ are the functionals described in the theorem above.

**Corollary 5.4.** For subspaces $W_1, W_2 \leq V$ with $\dim V < \infty$,

$$W_1 = W_2 \iff W_1^0 = W_2^0.$$

## 6 Properties of Annihilators

**Proposition 6.1.** Let $V$ be a vector space over $F$, and let $W_1, W_2 \leq V$ be subspaces. Then:

1. $(W_1 + W_2)^0 = W_1^0 \cap W_2^0$
2. $(W_1 \cap W_2)^0 = W_1^0 + W_2^0$

*Proof.* 1. is trivial. To show 2., let $f \in (W_1 \cap W_2)^0$. Define a map

$$f_1 : W_1 + W_2 \to F, \quad \alpha_1 + \alpha_2 \mapsto f(\alpha_2)$$

It is well-defined because $\alpha_1 + \alpha_2 = \alpha_1' + \alpha_2'$ implies $\alpha_2 - \alpha_2' = \alpha_1' - \alpha_1 \in W_1 \cap W_2$, and since $f \in (W_1 \cap W_2)^0$, we have

$$f(\alpha_2 - \alpha_2') = f(\alpha_2) - f(\alpha_2') = 0$$

which gives $f(\alpha_2) = f(\alpha_2')$.
Now, for a basis $\beta$ of $W_1 + W_2$ and an extension $\beta \cup \beta'$ for $V$, define an extension $\overline{f_1} : V \to F$ such that:

$$\overline{f_1}(\alpha) = \begin{cases} f_1(\alpha) & \texttt{if } \alpha \in \langle \beta \rangle \\ 0 & \texttt{if } \alpha \in \langle \beta' \rangle \end{cases}$$

Put $f_2 = f - \overline{f_1}$. Then $f = \overline{f_1} + f_2$, $\overline{f_1} \in W_1^0$, and $f_2 \in W_2^0$, thus $(W_1 \cap W_2)^0 \subseteq W_1^0 + W_2^0$. The converse is trivial. □

**Theorem 6.2.** Let $W \leq V$ and let $\{g_1, \ldots, g_r\}$ be a spanning set for $W^0$. Then

$$W = \bigcap_{i=1}^{r} \ker g_i$$

*Proof.* The inclusion $W \subseteq \bigcap_{i=1}^{r} \ker g_i$ is trivial, because by the definition of the annihilator, $W \subseteq \ker g$ for all $g \in W^0$.

Conversely, let $\{f_1, \ldots, f_m\} \subseteq V^*$ be a set of linear functionals such that $W = \bigcap_{i=1}^{m} \ker f_i$. Then for each $1 \leq i \leq m$,

$$W \subseteq \ker f_i \iff f_i \in W^0.$$

Since $\{g_1, \ldots, g_r\}$ is a spanning set for $W^0$, each $f_i$ is a linear combination of $\{g_1, \ldots, g_r\}$.
By the previously proved thm 4.2, this implies:

$$\bigcap_{j=1}^{r} \ker g_j \subseteq \ker f_i \quad \texttt{for each } i = 1, \ldots, m.$$

Taking the intersection over all $i$, we obtain:

$$\bigcap_{j=1}^{r} \ker g_j \subseteq \bigcap_{i=1}^{m} \ker f_i = W.$$

Thus, the equality is proved. □

## 7 Transpose and Annihilators

**Theorem 7.1.** Let $V$ and $W$ be vector spaces over $F$, and let $T : V \to W$ be a linear map. Then

$$\ker T^t = (\texttt{Im } T)^0.$$

In particular, if $V$ and $W$ are finite-dimensional, then:

1. rank $T^t =$ rank $T$
2. Im $T^t = (\ker T)^0$

*Proof.* The first assertion is proved as follows:

$$g \in \ker T^t \iff T^t(g) = 0 \iff g(T(\alpha)) = 0 \text{ for all } \alpha \in V \iff g(\gamma) = 0 \text{ for all } \gamma \in \texttt{Im } T \iff g \in (\texttt{Im } T)^0.$$

Thus, $\ker T^t = (\texttt{Im } T)^0$. Now, supppose that $V$ and $W$ have finite-dimensional.
Put $n = \dim V$ and $m = \dim W$ and $r = \operatorname{rank} T$. Since $\dim \operatorname{im} T + \dim(\operatorname{im} T)^0 = \dim W$, $\dim(\operatorname{im} T)^0 = m - r$.
Since the first result of this theorem with the Dimension Theorem,

$$m - r = \dim \ker T^t = \dim W^* - \dim \operatorname{im} T^t = m - \operatorname{rank} T^t$$

Therefore, $\operatorname{rank} T^t = m$. To show the last assertion, let $f \in \operatorname{im} T^t$. That is, $f = T^t(g)$ for some $g \in W^*$. Now,

$$f(\alpha) = T^t(g)(\alpha) = (g \circ T)(\alpha) = g(0) = 0 \text{ for all } \alpha \in \ker T$$

Thus $f \in (\ker T)^0$, or $\operatorname{im} T^t \subseteq (\ker T)^0$. Finally,

$$\dim(\ker T)^0 = \dim V^* - \dim \ker T = \operatorname{rank} T = \operatorname{rank} T^t$$

Proved. □

# 8 System of Linear Equations in the View of Linear functionals

Consider a system of linear equations:

$$\begin{cases} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n = 0 \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n = 0 \\ \quad\vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n = 0 \end{cases}$$

For each $1 \le i \le m$, define a linear functional $f_i : F^n \to F$ as:

$$f_i(x_1, \ldots, x_n) = \sum_{j=1}^{n} A_{ij}x_j$$

Then, solving the system of equations is equivalent to finding the subspace $W \subseteq F^n$:

$$W = \ker f_1 \cap \ker f_2 \cap \cdots \cap \ker f_m = \bigcap_{i=1}^{m} \ker f_i$$

Using the natural identification,

$$W = \bigcap_{i=1}^{m} \ker f_i = \left( \sum_{i=1}^{m} (\ker f_i)^0 \right)^0 = \left( \sum_{i=1}^{m} \langle f_i \rangle \right)^0 = \{f_1, \ldots, f_m\}^0$$

That is, the solution set is exactly the Annihilator of the row space.
Let $g_1, \ldots, g_r$ be a basis for $\langle f_1, \ldots, f_m \rangle$. Then clearly $r \le m$.
And, choose $g_{r+1}, \ldots, g_n \in V^*$ such that $g_1, \ldots, g_n$ be a basis for $V^*$. Now, there exists a basis $\{\alpha_1, \ldots, \alpha_n\}$ for $V$ s.t

$$g_i(\alpha_j) = \delta_{ij}$$

Finally, $\{\alpha_{r+1}, \ldots, \alpha_n\}$ be a basis for $W$.
Conversely, suppose that $W \le F^n$ is a subspace. How can we find the linear system having $W$ as a solution set?

# 9 Inner Product Space and Linear functional

Recall that:

> **Theorem 9.1.** Let $V$ be a finite dimensional inner product vector space over $F$.
> For any linear functional $g : V \to F$, there exists a unique vector $y \in V$ such that
>
> $$g(x) = \langle x, y \rangle, \;\; \forall x \in V$$